Ariel Bailon Intriago

AI Engineer · Full-Stack Engineer (backend) · Agents, RAG and evals on Node.js/NestJS and Python

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CAREER SUMMARY

Backend-focused Full-Stack Engineer shipping production Node.js/NestJS backends since 2020, now building LLM systems end to end: tool-calling agents, cited RAG, automated evals and observability. My latest project, OrderOps Agent, is a NestJS + LangGraph.js support agent where the model proposes and the code decides: refund rules are enforced in code and every refund goes through human approval. It runs on AWS with CI/CD, per-request tracing and a public demo (demo.arielbailon.dev), and its quality is gated by **41 eval cases** in CI. Production track record of cutting API response times (**~35%**) and load times (**~25%**). Looking for an AI Engineer or AI-focused backend role where I can take features from problem to production and measure whether they work.

TECHNICAL SKILLS

Agents & LLMs: LangGraph.js, LangChain.js, tool/function calling, human-in-the-loop, multi-turn memory, Anthropic Claude, OpenAI, AWS Bedrock, OpenRouter, provider failover, prompt caching, structured outputs (Zod, Pydantic), SSE streaming

RAG: embeddings (sentence-transformers), vector search (pgvector, ChromaDB), section-aware chunking, cited answers, refusal guardrails

Evaluation & observability: LLM-as-judge evals, adversarial cases, regression gate in CI, Langfuse tracing, per-request cost/token/latency tracking, Vitest, pytest

Backend: Node.js, NestJS, TypeScript, Python, FastAPI, PostgreSQL, MongoDB, MySQL, GraphQL, REST, Prisma, Redis, BullMQ

Frontend, infrastructure & CI/CD: React, Next.js, Angular · Docker, AWS (EC2, ECR, SSM, IAM, CloudWatch, S3, Bedrock), Caddy, NGINX, DigitalOcean · GitHub Actions (CI/CD with OIDC)

AI-assisted delivery: Claude Code with a spec-first workflow (spec → implement → test/type/lint checks → fix records); 31 specs and 15 fix records across 3 public repos

AI PROJECTS — END-TO-END OWNERSHIP

OrderOps Agent | demo.arielbailon.dev | github.com/ArielBailon/order-ops — Oct. 2026

E-commerce support agent that answers policy questions, looks up orders and proposes refunds a human approves — solo build, in production with a public demo

- Designed a LangGraph.js state graph with an intent router and **4 routes**: store policies (RAG over pgvector), order lookups (tool calling with the user's identity, so each customer only sees their own orders), refunds, and controlled out-of-scope refusal.
- Separated generative from deterministic logic: refund rules live in code, not the prompt, and the agent pauses (interrupt) until an operator approves or rejects each refund. Anthropic, OpenAI and AWS Bedrock behind one interface with provider failover and prompt caching.
- Deployed on AWS: EC2 behind Caddy (TLS) with Bedrock through an IAM instance role, no access keys on the server. CI/CD with GitHub Actions: OIDC, ECR images tagged by commit, deploy over SSM, smoke test and automatic rollback. The public demo has rate limits, a daily spend cap and a nightly reset.
- Observability: every request is traced in PostgreSQL and Langfuse (route, tools, provider, failover, tokens, cost, latency) and shown in a Next.js console with an agent timeline and a p50/p95 dashboard; **~$0.012** per conversation.
- **41 eval cases** (**13 adversarial**), **41/41** on the latest production run, used as a regression gate on every pull request; **457 automated tests**.

Technologies: TypeScript, NestJS, LangGraph.js, LangChain.js, PostgreSQL + pgvector, Prisma, Zod, Next.js, Langfuse, AWS (EC2, ECR, SSM, IAM, CloudWatch, Bedrock), Docker, GitHub Actions

EDGAR Analyst | github.com/ArielBailon/edgar-analyst — Sep. 2026

Grounded Q&A over SEC filings and earnings calls, citing the exact filing and section — solo build

- RAG pipeline over 10-K/10-Q filings and transcripts: section-aware chunking (**~2,400 cited chunks**), semantic search with ChromaDB and Claude Haiku 4.5 answers with exact citations.
- Pre-LLM refusal guardrail: no supporting evidence, no answer, so off-topic questions cost **$0**.
- Eval set of **22 hand-labeled pairs** (citation accuracy and relevance); diagnosed the gap as a retrieval problem, not a generation one, and documented the improvement hypotheses.
- Per-query logging of latency, tokens and cost (**~2 s** median, **~$0.002** per answer); **122 deterministic pytest tests**.

Technologies: Python, Anthropic API, ChromaDB, sentence-transformers, Pydantic, pytest

PROFESSIONAL EXPERIENCE

Independent — AI Engineering
Jul. 2026 – Present

Own AI projects: OrderOps Agent and EDGAR Analyst (above) and CodeSage

- **CodeSage (in progress):** assistant that answers questions about a code repository. FastAPI microservice with SSE streaming, Pydantic-validated output, exponential-backoff retries, per-request cost tracking and versioned prompts; **74 tests**. Next: RAG with pgvector, hybrid search and file-and-line citations.

Technologies: Python, FastAPI, Pydantic, PostgreSQL + pgvector, TypeScript, NestJS, Docker

ULTAHOST, United States
Aug. 2025 – Oct. 2025

Backend Developer (Contract)

- Designed and implemented scalable backend APIs with NestJS and TypeScript, improving response times by **~35%** through query optimization and caching.
- Built and maintained REST and GraphQL services integrated with internal and third-party systems.
- Implemented a microservices architecture for high-availability systems on MySQL.

Technologies: NestJS, TypeScript, MySQL, GraphQL

MUSHROOMSOFT-IT, Quito, Ecuador
Jun. 2023 – Jan. 2024

Full Stack Developer

- Developed backend services for a micro-frontend architecture (NestJS + Angular), cutting load times by **~25%** through PostgreSQL indexing and query optimization.
- Designed data models in PostgreSQL and MongoDB for a system with **50k+** active records.
- Containerized applications with Docker, standardizing environments and eliminating configuration incidents across the team.
- Led weekly code reviews, establishing quality standards adopted by a team of 5 developers.

Technologies: NestJS, Angular, PostgreSQL, MongoDB, Docker

ZGAMESLATAM, Manta, Ecuador
Jun. 2022 – Jun. 2023

Full Stack Developer

- Designed and implemented REST APIs in Node.js for an internal ERP processing **~200 daily transactions**, improving data consistency across modules.
- Optimized NGINX server configuration, achieving a **~20%** reduction in average latency.
- Managed deployments on AWS and DigitalOcean for a team of 4.

Technologies: Node.js, Angular, AWS, DigitalOcean, NGINX

CONSULTI, Guayaquil, Ecuador
Sep. 2020 – Jun. 2022

Junior Full Stack Developer

- Developed backend APIs in Node.js and Spring Boot for an e-commerce platform with **5,000+ products** and multiple integrated payment methods.
- Deployed applications on AWS and DigitalOcean using Docker and NGINX, reducing deploy time by **~40%**.
- Applied clean code and Scrum in a team of 6, consistently meeting delivery deadlines.

Technologies: Node.js, Spring Boot, Docker, AWS, NGINX

EDUCATION

Universidad Laica "Eloy Alfaro" de Manabí, Manta, Ecuador
2014 – 2021

BS, Systems Engineering

LANGUAGES

Spanish (native — C2) | English (advanced — C1)